

Normalized Prefix Search

Problem Statement

You are given a sorted array of strings N and a query string S .

Normalization Rules:

1. Convert every character to lowercase.
2. Remove every character that is not a letter or digit. Ignore spaces and all special characters.

Task:

Normalize every string in N and normalize S . Find all strings whose normalized form starts with the normalized query.

Return [count, firstIndex, lastIndex]. If no match exists, return [0, -1, -1].

Example:

N = [API Design Guide, api-design: basics, Apple Pie, App-le: pie!!, application security, banana]

S = app

Normalized strings: apidesignguide, apidesignbasics, applepie, applepie, applicationsecurity, banana

Output: [3, 2, 4]

Approach:

Normalize all strings using lowercase + keep only alphanumeric characters. Normalize the query. Since the normalized array is sorted, use binary search to find the first and last matching indices.